Ensemble Learning and Feature Selection for Automated ECG Arrhythmia Classification: A Comparative Analysis

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***Abstract*—Timely diagnosis of electrocardiogram (ECG) signals for arterial occlusions is important in preventing severe cardiac events like myocardial infarction. ECG diagnosis is difficult owing to the high-dimensionality, noise, and heterogeneity of biomedical signals. Single machine learning (ML) model output and feature selection are typically unstable or inconsistent and therefore lack clinical applicability. To tackle this, the current research suggests a hybrid ensemble learning approach that integrates prediction-level and feature-level aggregation to make the model more stable, interpretable, and predictive for occlusion detection. At the feature level, techniques like Mean Rank, Mean Weight, Robust Rank Aggregation (RRA), Threshold Algorithm (TA), and MedRank (MA) are experimented to obtain a stable list of features using different than one selection approach. At the predictive level, ensemble methods like hard voting, soft voting, weighted average, and stacking are utilized to merge outputs of different classifiers like CNN, Random Forest, XGBoost, and SVM. The resulting two-level ensemble architecture is evaluated on ECG datasets based on performance metrics like Accuracy, F1-score, and AUC as well as feature stability indices. This concurrent aggregation method is expected to enhance the clinical validity and consistency of ML-based occlusion detection with an eventual stepping stone to more stable and interpretable AI systems in biomedical diagnostics.**

I. Introduction

Cardiovascular diseases (CVDs) continue to be the leading cause of death worldwide, accounting for over one-third of all deaths each year [1]. One of the most dangerous and time-dependent diagnoses to make is myocardial infarction (heart attack) brought on by arterial blockage. To improve patient care and survival rates, occlusion must be detected quickly and urgently. This involves finding obstructions that decrease the amount of blood that reaches cardiac tissue.

The ECG (Electrocardiograms) is perhaps the most accessible and least invasive diagnostic technique for identifying such occlusions. The ECG measures the heart's electrical activity and gives temporal pattern outputs like P-waves, QRS complexes, and T-wave morphologies used to characterize underlying cardiac events. Yet, the manual interpretation of ECG signals is generally subjective and inaccurate, particularly in the instance of subtle or NSTEMI (Non-ST-Elevation Myocardial Infarction), a type of heart attack, when ECG alterations are not visually apparent [2]. Even computerized ECG interpretive software readily available does not identify such subtle occlusion signals owing to patient variation, noise, and co-existing overlapping pathophysiological conditions.

Developments in machine learning (ML) have been promising in automated analysis of ECG and enhanced diagnostic accuracy. ML algorithms are capable of learning subtle, non-linear relationships between signal characteristics and clinical annotations. Nevertheless, despite having accomplished this, a single ML model remains prone to instability i.e., small variations in training data, preprocessing, or feature extraction techniques can result in wildly divergent outcomes. This non-robustness decreases clinical reliability, which is required for medical use where both reliability and consistency are at least as valuable as predictive accuracy [3].

A. Problem Statement

Developing reliable machine learning (ML) models for biomedical signal analysis, particularly for electrocardiogram (ECG) classification, remains challenging due to the inherent complexity of the data. ECG signals are high-dimensional, noisy, and sample-sparse, often containing thousands of time-series data points but only a limited number of labeled patient samples. This imbalance increases the risk of overfitting, where models learn to memorize patterns in the training data rather than generalizing to unseen patients.

Beyond data scarcity, feature instability is another key concern. Different feature selection algorithms—such as statistical, filter-based, or model-based methods, often identify inconsistent sets of important features from the same dataset. This inconsistency undermines model reproducibility, both of which are critical in biomedical applications where clinicians require transparent and explainable decision-making.

At the prediction level, ensemble diversity can also lead to inconsistent outputs among classifiers trained on similar data. For instance, a convolutional neural network (CNN) may detect a cardiac occlusion, while a support vector machine (SVM) may fail to do so on the same input. Such prediction

variability leads to diagnostic uncertainty, limiting clinical trust in AI-assisted systems.

Therefore, the central problem this research addresses is the lack of stability and reproducibility in both feature selection and model predictions for ECG classification. To overcome this, the study focuses on developing a hybrid ensemble framework that integrates feature-level and prediction-level aggregation, aimed at enhancing robustness, and diagnostic consistency across biomedical ML models.

B. Proposed Direction

To address feature and prediction instabilities, this project proposes a hybrid ensemble learning framework integrating two aggregation levels. At the feature level, five methods combine multiple feature selection results into stable consensus subsets: Mean Rank (MR), Mean Weight (MW), Robust Rank Aggregation (RRA), Threshold Algorithm (TA), and MedRank (MA). Features consistently ranked high across methods are more biologically meaningful and robust to noise.

At the prediction level, four strategies combine outputs from 13 diverse classifiers (Random Forest, Gradient Boosting, AdaBoost, MLP, SVM variants, Naive Bayes, KNN, Decision Trees): Hard Voting (majority vote), Soft Voting (averaged probabilities), Weighted Averaging (performance-based weighting), and Stacking (meta-learner optimization). This diversity leverages complementary strengths across linear, non-linear, and probabilistic models.

The objective is to rigorously evaluate this two-level framework using Accuracy, F1-score, AUC, and stability metrics, producing an ensemble that is accurate, consistent, and clinically reliable for real-world cardiac diagnosis.

II. Background

Over the past decade, there has been extensive growth in the incorporation of machine learning (ML) techniques in the processing of biomedical signals to interpret electrocardiogram (ECG) signals. This is because ML-based diagnostic models have shown a high level of accuracy in detecting intricacies associated with heart patterns that cannot be easily identified through direct human interpretation. Models such as convolutional neural networks (CNNs), support vector machine (SVM) models, and gradient boosting models have shown high accuracy in correctly identifying heart disorders. But although they have high accuracy in predictive modeling tasks, they have been less reliable in terms of reproducibility.

This instability is shown at both the feature level and the predictive level. At the feature level, a noisy, heterogeneous, and high-dimensional nature of biomedical data such as ECGs causes inconsistent results among various feature selection methods or even within the same method at cross-validation iterations. This directly imparts little biological plausibility to the model because it keeps inventing new features to be given high importance. At the predictive level, results among similar data partitions can differ because a model is not driven decisively by underlying patterns in the data. This imparts inconsistency to ML-based results in the field of biomedical diagnostics.

To counter the instability in the process of feature selection, there have been a few statistical methods of aggregation that have been developed. For example, Robust Rank Aggregation (RRA) is one technique that uses a probabilistic model to determine those features that have been top-ranked in many lists. This is done to filter out noise and determine those features that are of real importance. Other methods like MedRank and Threshold Algorithms (TA) aggregate the rankings of features to achieve a consensus [4]. These methods have proved to be successful in genomics and biomarker discovery. But little work has been done in this area in ECG-based applications.

On the other hand, ensemble learning methods have been developed in parallel to improve the stabilization of predictions by means of combining several classifiers using methods such as bagging, boosting, voting, or stacking. It has been verified in the literature that ensemble learning can improve generalization with regard to predictions because it can improve the variance associated with certain tasks in the biomedical field. It is important to note that there is no method among those mentioned above that can improve the instability of features at the level of the features themselves. The lack of frameworks that allow for the integration of feature-level aggregation and prediction-level ensemble methods is a gap that is not addressed in current state-of-the-art research. Current studies mainly focus on improving models with regard to accuracy or AUC measures. But it is important to consider the concept of model stability because it is one of the paramount objectives when it comes to selecting models to be used in real-world experiments. It is with this background that this paper aims to bridge this gap with the proposed method.

III. Related Work

A. Ensemble Learning for ECG Prediction

In recent years, ensemble learning has become an increasingly prominent approach for electrocardiogram (ECG) classification tasks, because of its ability to decrease variance and improve robustness in accurately identifying arrhythmia. For example, research using a convolutional neural network (CNN) ensemble, as well as a combination of different deep learning architectures such as CNN + Bi-LSTM layers, has been effective in accurately identifying arrhythmia between multiple classifications. Another research study proposed a two-level deep learning ensemble system that utilized several different architectures of a neural network that was refined for identification of ECG spectrograms with an accuracy of over 95% in their testing data. [4] Another investigation used an ensemble of depth-wise separable CNNs for heartbeat classification and found improvements in both sensitivity and specificity compared to a single model baseline. [5]

In addition to architecture-level ensembles, techniques of prediction aggregation have been crucial in determining how predictions from ensembles will be aggregated. Traditional techniques within this field include majority voting, in which classifiers essentially get a vote to predict a particular output

and the output with the most votes wins, softmax average predictions in which predictions are assessed based upon their probabilities of being accurate and selecting the one that is highest when all probabilities are averaged, and stacking, whereby a meta-model is developed that learns based upon predictions of its component models. In ECG, considerations of complexity and variability in signals such as noises and individual differences in morphologies require robust methods of aggregation that would be effective in edge conditions with low-probability arrhythmias or noises in leads.

B. Rank Aggregation in Biomedical Applications

Feature selection and ranking play a critical role in biomedical data analysis, where datasets are often high-dimensional, noisy, and subject to variability across cohorts. In this respect, Robust Rank Aggregation (RRA) and MedRank have been utilized in biomedical research for multiple ranked-list integration, such as gene lists or biomarkers, in order to identify variables that were ranked higher than random expectation. [6] In a specific biomedical application, an ensemble feature selection procedure was applied to data-driven thresholding of Robust Rank Aggregation (RRA) and MedRank for Alzheimer's disease biomarker discovery. In this work, it was found that selection stability was increased up to 34%. [7]

Despite this, there is a gap in the literature: while many studies investigate ensemble feature selection and rank aggregation of the features, few explore how these feature-rank aggregation methods are integrated into model-level ensembles and prediction aggregation pipelines. In other words, whereas feature-rank consolidation has been well investigated in the field of genomics and biomarker discovery, its translation to full end-to-end pipelines, such as feature ranking → aggregation → model ensemble → prediction aggregation, remains limited in biomedical signal-processing domains like ECG. This project will fill this gap by proposing a combination of feature-rank aggregation (via methods such as mean rank, RRA, MedRank) with prediction-level ensemble strategies, namely voting, soft averaging, and stacking, within an ECG arrhythmia classification framework. In other words, investigating both levels of ensembles - feature and model - jointly.

IV. Methodology

A. Dataset

This study utilizes a large-scale 12-lead electrocardiogram (ECG) dataset originally shared on Figshare and subsequently made available on PhysioNet. [8] The dataset comprises 10,646 patient records, each containing a complete 12-lead ECG recording along with comprehensive clinical metadata and diagnostic annotations.

1) ECG Signal Data: Each patient record includes a 12-lead ECG signal captured using the GE MUSE system *(a system from GE HealthCare that integrates, manages, and streamlines the acquisition, storage, analysis, and distribution of cardiac-related data, primarily electrocardiograms (ECGs))*. The ECG recordings are stored as individual CSV files with the following characteristics:

1) **Sampling Rate:** 500 Hz (standard clinical ECG sampling frequency)
2) **Recording Length:** Approximately 10 seconds (4,999 samples per lead)
3) **Lead Configuration:** Standard 12-lead ECG including:
   a) Limb leads: I, II, III
   b) Augmented limb leads: aVR, aVL, aVF
   c) Precordial (chest) leads: V1, V2, V3, V4, V5, V6

The dataset provides both raw and denoised versions of the ECG signals. For this study, we utilized the denoised ECG recordings to minimize the impact of noise artifacts and baseline wander on feature extraction.

2) Clinical Metadata: In addition to the ECG signals, each patient record includes the following clinical and diagnostic information stored in a master diagnostics file (Diagnostics.xlsx):

- Patient Demographics:
  - Age (ranging from 4 to 98 years, mean age: 59.2 years)
  - Gender (55.9% male, 44.1% female)
- Cardiac Rhythm Classification:
  - Primary rhythm diagnosis (11 distinct rhythm types)
  - Beat characteristics and abnormalities
- ECG Summary Parameters (automatically extracted by GE MUSE):
  - Ventricular Rate (beats per minute)
  - Atrial Rate (beats per minute)
  - QRS Duration (milliseconds)
  - QT Interval (milliseconds)
  - QT Corrected (QTc, milliseconds)
  - R Axis (degrees)
  - T Axis (degrees)
  - QRS Count
  - Q Onset, Q Offset, T Offset (milliseconds)

3) Rhythm Distribution: The dataset contains 11 distinct cardiac rhythm types with the distribution shown in Table I

B. Data Preprocessing

The preprocessing pipeline consisted of four main stages: label creation, feature extraction, data cleaning, and train-validation-test splitting.

1) Binary Label Creation: For this arrhythmia classification task, we converted the 11 distinct rhythm types into a binary classification problem:

- **Normal (Class 0):** All sinus rhythm variants, representing normal cardiac electrical activity
  - Sinus Rhythm (SR)
  - Sinus Bradycardia (SB)
  - Sinus Tachycardia (ST)
  - Sinus Irregularity (SI)
  - Sinus Arrhythmia (SA)
  - Sinus Atrium to Atrial Wandering Rhythm (SAAWR)
- **Arrhythmia (Class 1):** Non-sinus rhythms indicating pathological cardiac conditions

TABLE I
DISTRIBUTION OF CARDIAC RHYTHM TYPES IN THE DATASET

Rhythm Type	Acronym	Count	Percentage
Sinus Bradycardia	SB	3,889	36.5%
Sinus Rhythm	SR	1,826	17.2%
Atrial Fibrillation	AFIB	1,780	16.7%
Sinus Tachycardia	ST	1,568	14.7%
Supraventricular Tachycardia	SVT	587	5.5%
Atrial Flutter	AF	445	4.2%
Sinus Arrhythmia	SA	399	3.7%
Atrial Tachycardia	AT	121	1.1%
AV Node Reentrant Tachycardia	AVNRT	16	0.2%
AV Reentrant Tachycardia	AVRT	8	0.1%
Sinus Atrium Wandering Rhythm	SAAWR	7	0.1%
Total		**10,646**	**100.0%**

- Atrial Fibrillation (AFIB)
- Atrial Flutter (AF)
- Supraventricular Tachycardia (SVT)
- Atrial Tachycardia (AT)
- Atrioventricular Node Reentrant Tachycardia (AVNRT)
- Atrioventricular Reentrant Tachycardia (AVRT)

This binary classification scheme resulted in the following class distribution:

- **Normal:** 7,682 patients (72.2%)
- **Arrhythmia:** 2,964 patients (27.8%)

2) Feature Extraction from ECG Signals:: Given the computational challenges of processing raw ECG time-series data (approximately 60,000 data points per patient), we extracted a comprehensive set of statistical and morphological features from each 12-lead ECG recording. This approach reduces dimensionality while preserving clinically relevant information. For each of the 12 ECG leads, we computed the following 13 features, resulting in 156 ECG-derived features per patient:

Statistical Features:

- Mean: Average signal amplitude
- Standard Deviation: Signal variability
- Minimum and Maximum: Amplitude range extremes
- Median: Central tendency measure
- Range: Peak-to-peak amplitude

Distribution Features:

- 25th Percentile (Q1): Lower quartile
- 75th Percentile (Q3): Upper quartile
- Interquartile Range (IQR): Q3 - Q1
- Skewness: Distribution asymmetry
- Kurtosis: Distribution tailedness

Energy Features:

- Root Mean Square (RMS): Signal power
- Total Energy: Sum of squared amplitudes

These features capture both the overall characteristics of the ECG waveform and its statistical properties, which are known to differ between normal and arrhythmic cardiac patterns.

3) Clinical Metadata Integration: In addition to the 156 ECG-derived features, we incorporated 7 clinical metadata features directly from the GE MUSE system diagnostics:

1) Patient Age: Numerical age in years
2) Gender: Encoded as binary (Male = 1, Female = 0)
3) Ventricular Rate: Heart rate measured in beats per minute
4) Atrial Rate: Atrial contraction rate in beats per minute
5) QRS Duration: Duration of ventricular depolarization (ms)
6) QT Interval: Time from ventricular depolarization to repolarization (ms)
7) QT Corrected (QTc): Heart rate-corrected QT interval (ms)

These clinical parameters are routinely used in clinical ECG interpretation and provide complementary information to the signal-derived features. The final feature set contained 163 features per patient (156 ECG features + 7 clinical metadata features).

4) Data Quality Control: Before model training, we performed data quality checks and cleaning:

1) Missing Value Handling: The dataset contained no missing values in either the ECG recordings or clinical metadata.
2) Categorical Encoding: Gender was encoded numerically (Male = 1, Female = 0) to enable use in machine learning algorithms.
3) Infinite Value Check: We verified that no infinite values were present in the extracted features.
4) Feature Normalization: All features were standardized using StandardScaler to have zero mean and unit variance. This normalization was performed on the training set only, and the same transformation parameters were applied to validation and test sets to prevent data leakage.

5) Dataset Splitting: The preprocessed dataset was split into three subsets using stratified sampling to maintain class distribution across all splits:

- **Training Set:** 72.0% (6,916 patients)
  - Normal: 4,989 patients (72.2%)
  - Arrhythmia: 1,927 patients (27.8%)
- **Validation Set:** 8.0% (851 patients)
  - Normal: 615 patients (72.2%)
  - Arrhythmia: 236 patients (27.8%)

- **Test Set:** 20.0% (1,879 patients)
  - Normal: 1,078 patients (72.2%)
  - Arrhythmia: 801 patients (27.8%)

6) Data Storage: The final preprocessed dataset was saved in CSV format with the following structure:

- **Rows:** Individual patient records (10,646 total)
- **Columns:**
  - 156 ECG-derived features (named as Lead_Feature, e.g., I_mean, V1_std)
  - 7 clinical metadata features
  - 1 target label (0 = Normal, 1 = Arrhythmia)
  - 2 identifier columns (FileName, Rhythm)

Additionally, separate CSV files were created for training, validation, and test sets to facilitate model development and evaluation. The StandardScaler parameters were saved as a serialized file (scaler.pkl) to ensure consistent feature normalization during model deployment.

C. Feature Selection

High-dimensional biomedical data presents significant challenges for machine learning models. In this study, each patient record initially contained 163 features comprising 156 ECG-derived statistical features and 7 clinical metadata features. While comprehensive, this high dimensionality introduces several critical issues that necessitate feature selection. With limited patient samples (10,646 records) relative to the feature space, models are susceptible to the curse of dimensionality, becoming prone to overfitting where they memorize training-specific patterns rather than generalizing to unseen patients.

Training multiple classifiers on the full feature set is computationally expensive, particularly for ensemble methods that require numerous models across different cross-validation folds. Clinical applications demand interpretable models where healthcare professionals can understand which cardiac characteristics drive diagnostic decisions, making dimensionality reduction essential for practical deployment. Many ECG-derived features exhibit high correlation, such as mean and median values from the same lead, introducing redundancy that can dilute model performance.

Furthermore, not all extracted features contribute meaningfully to arrhythmia detection; some may capture measurement noise or patient-specific variations irrelevant to the diagnostic task. Most critically, feature selection instability poses a fundamental challenge: different feature selection algorithms applied to the same dataset often produce divergent rankings and subsets, undermining reproducibility and clinical reliability. This inconsistency necessitates a robust aggregation approach to identify features with genuine biological significance rather than method-specific artifacts.

To address these challenges, five distinct feature selection algorithms were used, each capturing different aspects of feature importance:

1) ***Random Forest Feature Importance:*** Random Forest (RF) is an ensemble of decision trees that naturally provides feature importance scores based on how much each feature decreases impurity (Gini index or entropy) across all trees in the forest [4]. Features that consistently appear in split nodes near the tree roots and lead to substantial impurity reduction receive higher importance scores.

Why used: RF importance captures non-linear relationships and feature interactions, making it well-suited for complex biomedical data. It is robust to outliers and handles mixed feature types (continuous and categorical) effectively.

2) ***Mutual Information (MI):*** Mutual Information measures the dependency between each feature and the target variable, quantifying how much information about the class label is gained by observing a particular feature [5]. It is calculated as:

$$I(X;Y) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} p(x,y) \, \log\left(\frac{p(x,y)}{p(x)\,p(y)}\right)$$

where X is the feature, Y is the class label, and p denotes probability distributions.

Why used: MI is model-agnostic and captures both linear and non-linear dependencies without assuming specific functional relationships. It detects features that are informative regardless of the classification algorithm employed.

3) ***Chi-Square (X^2) Test:*** The Chi-Square test evaluates the independence between each feature and the target class by comparing observed and expected frequency distributions [6]. Higher X^2 scores indicate stronger dependency between the feature and class label. Since X^2 requires non-negative values, we applied absolute value transformation to the feature matrix.

Why used: Chi-Square is particularly effective for categorical or discretized features and provides statistical significance testing. It identifies features whose distributions differ significantly between normal and arrhythmia classes.

4) ***ANOVA F-value:*** Analysis of Variance (ANOVA) F-test assesses whether feature means differ significantly across classes [7]. The F-statistic is computed as the ratio of between-class variance to within-class variance:

$$F = \frac{\sigma^2_{\text{between}}}{\sigma^2_{\text{within}}}$$

Higher F-values indicate that the feature exhibits greater separation between classes relative to variation within classes.

Why used: ANOVA F-test is appropriate for continuous features and assumes normally distributed data. It effectively identifies features where class separation is pronounced, making them discriminative for classification.

5) ***Recursive Feature Elimination (RFE) with SVM:*** RFE is a wrapper-based method that recursively removes the least important features based on model coefficients [8]. We employed Support Vector Machine (SVM) with a linear kernel as the base estimator, iteratively eliminating 20% of features per step until reaching a target subset size. SVM assigns weights to features based on their contribution to the maximum-margin hyperplane.

Why used: Unlike filter methods (RF, MI, X^2, ANOVA) that evaluate features independently, RFE considers feature interactions and their combined effect on model performance.

This provides a complementary perspective grounded in actual classifier behavior.

D. Feature Aggregation

Given that different feature selection methods produce divergent rankings—a phenomenon well-documented in biomedical machine learning [9]—we employed five aggregation strategies to combine these rankings into stable, consensus feature subsets:

*1) **Mean Rank (MR) Aggregation**:* Mean Rank aggregation assigns each feature a rank position within each selection method's output (1 for most important, N for least important) and computes the average rank across all methods:

$$\text{MeanRank}(f) = \frac{1}{M}\sum_{i=1}^{M}\text{Rank}_i(f)$$

where M is the number of feature selection methods and Rank_i(f) is the rank of feature f in method i. Features are then sorted by ascending mean rank, with lower values indicating higher consensus importance. This approach treats all selection methods equally and provides a simple, interpretable aggregation.

Rationale: Mean Rank is robust to outliers in importance scores (since it uses ordinal ranks rather than cardinal values) and provides a democratic aggregation where each method contributes equally [10].

*2) **Mean Weight (MW) Aggregation**:* Rather than using ranks, Mean Weight aggregation directly averages the raw importance scores from each feature selection method:

$$\text{MeanWeight}(f) = \frac{1}{M}\sum_{i=1}^{M}\text{Score}_i(f)$$

where Score_i(f) is the importance score assigned to feature f by method i.

Features are sorted by descending mean weight, with higher values indicating greater importance.

Rationale: Mean Weight preserves magnitude information, allowing features with exceptionally high scores in one method to retain influence even if moderately ranked in others. This can capture "specialist" features that are critical according to specific algorithms.

*3) **Robust Rank Aggregation (RRA)**:* Robust Rank Aggregation is a statistically principled method developed for integrating ranked lists in genomics and biomarker discovery [11]. RRA evaluates whether a feature's rank is significantly better than expected by random chance under a null hypothesis.

For each feature f, RRA computes a p-value based on the beta distribution, representing the probability of observing its ranks (or better) across methods purely by chance:

$$\beta_{k,n}(x) = \sum_{\ell=k}^{n}\binom{n}{\ell}x^{\ell}(1-x)^{n-\ell}$$

n: total number of ranked lists; k: rank threshold; ℓ: summation index; $\binom{n}{\ell}$: binomial coefficient; x: rank quantile; $x^{\ell}(1-x)^{n-\ell}$: probability term under the null model.

$$\rho(r) = \min_{k=1,\ldots,n}\beta_{k,n}(r(k))$$

$r(k)$: normalized rank of the item at position k; $\beta_{k,n}(r(k))$: null-model probability of observing such a rank; $\rho(r)$: final aggregated score obtained by taking the minimum probability across all k.

Features with lower p-values are considered more robustly important. We normalized ranks to percentiles before p-value computation to account for varying list lengths.

Rationale: RRA provides statistical rigor by filtering out features that rank high in only one or two methods due to noise or method-specific biases. It identifies features that consistently appear near the top across diverse ranking schemes, indicating genuine biological relevance.

*4) **Threshold Algorithm (TA)**:* The Threshold Algorithm is a dynamic aggregation approach adapted from database middleware research [12]. It iteratively combines ranked lists by setting a score threshold and selecting features that exceed this threshold across multiple methods.

Our implementation:

1) Convert raw scores to standardized ranks
2) Compute mean score for each feature across methods
3) Sort features by descending mean score
4) Set threshold at the k-th feature's mean score
5) Select all features with mean scores greater than or equal to the threshold

The parameter k (target number of features) was set to the average subset size across individual methods, ensuring comparable dimensionality.

Rationale: TA provides a balance between top-k selection and consensus-based filtering. It dynamically adjusts to the data distribution and stops when a stable set of highly-ranked features is identified.

*5) **MedRank (Median Rank)**:* MedRank selects features based on their frequency of appearance across ranked lists. A feature is included in the final subset only if it appears in at least a specified proportion (e.g., 20%, 50%, 80%) of the input ranking lists.

$$\text{MedRank}(f,\theta) = \left\{ f \;\middle|\; \frac{\text{count}(f \text{ appears in top-}k)}{M} \geq \theta \right\}$$

where ϑ is the threshold proportion (e.g., 0.2 for 20%) and M is the number of methods.

Rationale: MedRank emphasizes consensus—features must demonstrate consistent importance across multiple diverse methods to be selected. This conservative approach reduces the risk of including method-specific artifacts while ensuring that selected features are robustly supported by multiple perspectives [13].

Parameter Selection: We used ϑ = 0.2 (20% presence requirement), meaning a feature must appear in at least one of the five methods' top-ranked lists to be considered. This relatively permissive threshold balances inclusivity with noise filtering.

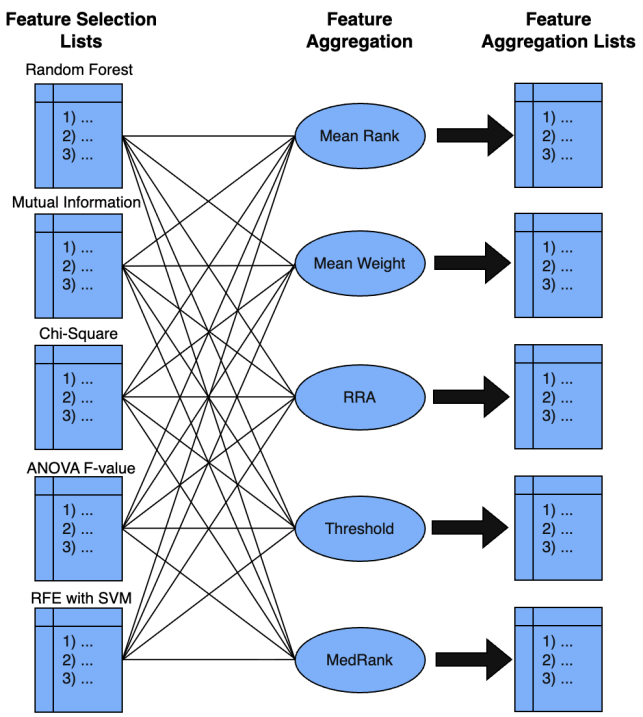

Fig. 1. Aggregating every list from feature selection to get final list.

Using five feature selection methods and five aggregation strategies may seem redundant, but it is intentionally designed to ensure robustness. Since no single algorithm consistently performs best across all datasets, each method contributes a unique perspective—RF captures feature interactions, MI detects nonlinear relationships, RFE models multivariate effects, and statistical tests like χ^2 and ANOVA provide strong theoretical grounding. By combining these complementary viewpoints through multiple aggregation approaches, the system achieves a more stable and reliable consensus on truly important ECG features.

E. Model Selection

Prediction-level variability is a major challenge in biomedical ML because different classifiers make different assumptions and capture different aspects of the data. Linear models rely on hyperplane separability, tree-based methods partition the feature space, and probabilistic models assume specific distributions—leading each to excel in some ECG patterns while missing others. Single models are also prone to overfitting and sensitive to data perturbations, which undermines reproducibility in clinical settings. Prediction aggregation mitigates these issues by combining outputs from multiple classifiers, leveraging their complementary strengths to produce more stable and reliable diagnostic decisions, similar to consulting multiple clinical experts.

1) Base Classifier Selection: We trained 13 diverse classifiers representing distinct learning paradigms to ensure comprehensive coverage of the algorithmic landscape. This diversity is essential for effective ensembling—if all models employ similar mechanisms, aggregation provides minimal benefit [4]. Our classifier suite includes:

1) **Linear Models:**
   - **Linear Discriminant Analysis (LDA):** LDA seeks a linear combination of features that maximally separates classes by maximizing the ratio of between-class variance to within-class variance [5]. It assumes normally distributed features with equal covariance across classes. LDA provides a simple, interpretable baseline with strong performance when linear separability exists. Its probabilistic output (via Bayes' theorem) supports soft voting.
   - **Quadratic Discriminant Analysis (QDA):** QDA relaxes LDA's assumption of equal class covariances, allowing each class to have its own covariance matrix, resulting in quadratic decision boundaries [5]. QDA handles cases where arrhythmia and normal classes exhibit different feature correlations, providing more flexible modeling than LDA while remaining interpretable.
2) **Probabilistic Models:**
   - **Gaussian Naive Bayes (GNB):** GNB applies Bayes' theorem with the "naive" assumption of conditional feature independence given the class label [6]. It models each feature's class-conditional distribution as Gaussian. Despite its strong independence assumption (often violated in ECG data), GNB frequently performs surprisingly well due to its low variance and robustness to irrelevant features.
   - **Bernoulli Naive Bayes (BNB):** NB is designed for binary/boolean features, modeling each feature as a Bernoulli distribution [6]. While our features are continuous, BNB can be applied after binarization thresholds. It provides a complementary probabilistic perspective and is extremely fast.
3) **Instance-Based Learning:**
   - **K-Nearest Neighbors (KNN):** KNN is a non-parametric method that classifies samples based on the majority class among their k nearest neighbors in feature space [7]. Distance is typically measured using Euclidean or Manhattan metrics. KNN makes no distributional assumptions and naturally captures local patterns. It excels when similar ECG profiles indicate similar diagnoses, reflecting clinical reasoning based on pattern recognition.
4) **Tree-Based Models:**
   - **Decision Tree Classifier (DTC):** Decision trees recursively partition the feature space using binary splits that maximize information gain or minimize Gini impurity [8]. Each leaf node represents a class prediction. Trees are highly interpretable—decision paths directly correspond to clinical decision rules (e.g., "if QRS duration ¿ 120ms AND QTc ¿ 450ms,

then arrhythmia"). They handle non-linear relationships and feature interactions naturally.

- **Extra Tree Classifier (ETC):** Extra Trees (Extremely Randomized Trees) are similar to decision trees but use random thresholds for splits rather than optimal ones, reducing variance at the cost of slight bias increase [9]. ETC's randomness makes it more robust to overfitting than standard decision trees while maintaining interpretability.

5) **Ensemble Tree Methods:**

- **Random Forest (RF):** Random Forest constructs multiple decision trees on bootstrapped samples of data, with each split considering only a random subset of features [10]. Predictions are aggregated via majority voting (classification) or averaging (regression). RF is one of the most robust and widely-used algorithms in biomedical applications. It handles high dimensionality, provides feature importance measures, and is resistant to overfitting.
- **Extra Trees Ensemble:** This is an ensemble of Extra Tree Classifiers, combining the randomness of individual Extra Trees with the variance reduction of ensemble aggregation [9]. Extra Trees Ensemble often achieves accuracy comparable to Random Forest with significantly faster training due to random split selection.
- **Gradient Boosting Classifier:** Gradient Boosting builds trees sequentially, with each tree correcting errors made by the previous ensemble [11]. It optimizes a loss function via gradient descent in function space. Gradient Boosting often achieves state-of-the-art performance on tabular data by iteratively focusing on hard-to-classify samples. It excels at capturing complex non-linear patterns.
- **AdaBoost (Adaptive Boosting):** AdaBoost trains weak learners (typically shallow decision trees) sequentially, with each learner weighted by its accuracy and training samples reweighted to emphasize misclassified instances [12]. AdaBoost adaptively focuses on difficult cases, making it effective for imbalanced datasets like ours (72.2% normal vs. 27.8% arrhythmia).
- **Bagging Classifier:** Bagging (Bootstrap Aggregating) trains multiple instances of the same base classifier on different bootstrap samples of the training data, then aggregates predictions via voting [13]. Bagging reduces variance and overfitting, particularly for high-variance base learners. We used Decision Trees as the base estimator.

6) **Neural Networks:**

- **Multi-Layer Perceptron (MLP):** MLP is a feed-forward artificial neural network with one or more hidden layers of neurons using non-linear activation functions (e.g., ReLU, tanh) [14]. It learns representations through backpropagation. MLPs can approximate arbitrary non-linear functions and learn complex feature interactions automatically. They represent the neural network perspective in our ensemble, complementing traditional ML approaches.

2) Training Protocol:

- Feature Normalization: All features were standardized using z-score normalization (StandardScaler) to have zero mean and unit variance:

$$x_{\text{normalized}} = \frac{x - \mu}{\sigma}$$

  where μ is the feature mean and σ is the standard deviation, computed only on the training set to prevent data leakage. The same transformation parameters were applied to validation and test sets.
- Training Configuration: For each of the 5 aggregated feature subsets (from Mean Rank, Mean Weight, RRA, Threshold Algorithm, MedRank) plus the baseline (all 163 features), we trained all 13 classifiers. This yielded 6 × 13 = **78 trained models**.

  Hyperparameter Settings:
  - **Random Forest:** 100 trees, Gini impurity, sqrt(n_features) per split.
  - **Gradient Boosting:** 100 estimators, learning rate = 0.1, max depth = 3.
  - **AdaBoost:** 100 estimators, learning rate = 1.0
  - **Bagging:** 100 estimators, bootstrap = True.
  - **MLP:** 2 hidden layers (100, 50 neurons), ReLU activation, Adam optimizer, max 500 iterations.
  - **KNN:** k = 5 neighbors, Euclidean distance.
  - **Decision Trees:** Gini impurity, no max depth (pruned via min_samples_split = 2).
  - **All others:** Default scikit-learn parameters.

  These settings represent standard, well-established configurations rather than exhaustive hyperparameter optimization. Our focus is on demonstrating ensemble aggregation benefits rather than maximum individual model performance.

F. Prediction Aggregation

Once all base classifiers are trained, prediction aggregation combines their outputs to produce final diagnostic decisions. We implemented four aggregation strategies, each with distinct characteristics and theoretical foundations.

- **Hard Voting (Majority Voting)** Hard voting treats each classifier as casting a discrete vote for one class. The final prediction is the class that receives the most votes:

$$\hat{y} = \text{mode}\{\, h_1(x),\, h_2(x),\, \ldots,\, h_n(x) \,\}$$

  where h_i(x) is the class prediction from classifier i, and mode selects the most frequent value.
- **Soft Voting (Probability Averaging)** Soft voting aggregates predicted class probabilities rather than discrete labels. For each class, probabilities from all classifiers are averaged, and the class with highest average probability is selected:

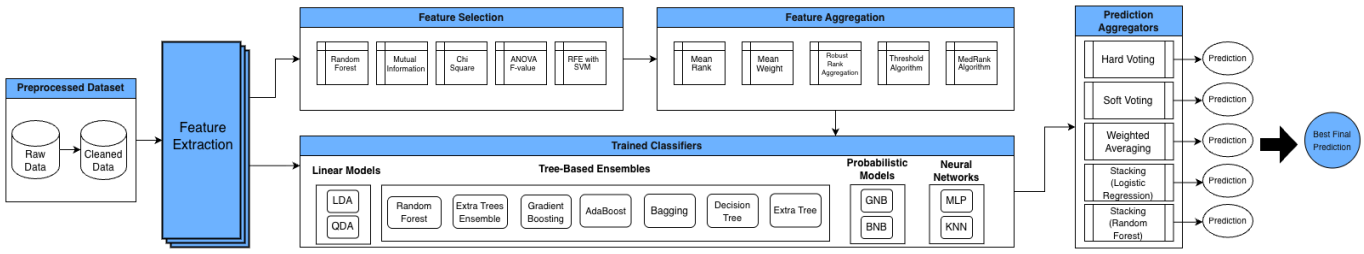

Fig. 2. Architecture

$$P(y = c \mid x) = \frac{1}{N} \sum_{i=1}^{N} P_i(y = c \mid x)$$

$$\hat{y} = \arg\max_{c} \; P(y = c \mid x)$$

$$P_i(y = c \mid x)$$

is the probability that classifier i assigns to class c for sample x.

- **Weighted Averaging** Weighted averaging extends soft voting by assigning each classifier a weight proportional to its validation performance. Better models receive more influence:

$$P(y = c \mid x) = \sum_{i=1}^{N} w_i \, P_i(y = c \mid x)$$

$$\sum_{i=1}^{N} w_i = 1$$

$$\hat{y} = \arg\max_{c} \; P(y = c \mid x)$$

Weight Calculation: We used validation accuracy as the performance metric for weight assignment:

$$w_i = \frac{\text{accuracy}_i}{\sum_{j=1}^{N} \text{accuracy}_j}$$

Each classifier was evaluated on the held-out validation set (851 patients), and its accuracy determined its weight contribution to the ensemble.

- **Stacking (Stacked Generalization)** Stacking is a two-level ensemble architecture where Level 0 (base models) generate predictions that become features for Level 1 (meta-learner), which learns the optimal combination strategy.
  - **Logistic Regression Meta-Learner:** Logistic regression learns linear combinations of base model predictions:

$$P(y = 1 \mid x) = \sigma\begin{pmatrix} \beta_0 + \beta_1 P_1(x) + \beta_2 P_2(x) \\ + \cdots + \beta_n P_n(x) \end{pmatrix}$$

where σ is the sigmoid function and β_i are learned weights.

  - **Random Forest Meta-Learner:** Random Forest can learn non-linear combinations and interactions:

$$\hat{y} = \text{RandomForest}\big(\big[P_1(x),\, P_2(x),\, \ldots,\, P_n(x)\big]\big)$$

Stacking is grounded in the theory of stacked generalization [20], which shows that a meta-learner can learn to correct systematic biases and exploit the complementary strengths of base models. It has theoretical connections to boosting, but with greater flexibility.

V. Results & Evaluations

The proposed hybrid ensemble learning framework was evaluated on 10,646 patient ECG records, partitioned into training (72%), validation (8%), and test (20%) sets with stratified sampling to maintain class distribution (72.2% Normal, 27.8% Arrhythmia). A total of 78 models were trained across 6 feature configurations (5 aggregated feature subsets plus baseline) and 13 classifiers, followed by evaluation of 5 ensemble prediction aggregation strategies.

A. Feature Aggregation Results

Table II summarizes the feature selection outcomes from each aggregation method. All aggregation methods selected 100 features (38.7% reduction from the original 163 features), achieving substantial dimensionality reduction while preserving diagnostic information.

TABLE II
Feature Aggregation Summary

Method	Features	Top-Ranked Feature	2nd Ranked Feature
Mean Rank	100	Atrial Rate	Ventricular Rate
Mean Weight	100	Atrial Rate	Ventricular Rate
RRA	100	aVR_std	aVR_median
Threshold	100	aVL_max	Gender
MedRank	100	aVR_std	V3_energy
Baseline	163	N/A (all features)	N/A

Despite employing different statistical and algorithmic approaches, several features consistently appeared in top rankings across multiple methods:

TABLE III
TOP 10 MODELS ON TEST SET

Rank	Aggregation Method	Classifier	Accuracy	F1-Score	Precision	Recall	Specificity
1	RRA	Gradient Boosting	94.55%	94.53%	94.52%	94.55%	96.81%
2	All Features	Gradient Boosting	94.46%	94.42%	94.42%	94.46%	96.94%
3	Mean Weight	Gradient Boosting	94.46%	94.42%	94.42%	94.46%	96.94%
4	Mean Rank	Gradient Boosting	94.23%	94.21%	94.20%	94.23%	96.36%
5	Mean Rank	Random Forest	94.04%	94.03%	94.03%	94.04%	95.97%
6	All Features	Random Forest	93.99%	93.96%	93.94%	93.99%	96.55%
7	Mean Weight	Random Forest	93.94%	93.91%	93.90%	93.94%	96.49%
8	RRA	Random Forest	93.71%	93.69%	93.68%	93.71%	95.97%
9	RRA	MLP	93.62%	93.62%	93.62%	93.62%	95.51%
10	Mean Rank	Extra Trees	93.52%	93.48%	93.47%	93.52%	96.23%

- Clinical Metadata: Atrial Rate and Ventricular Rate emerged as top features in both Mean Rank and Mean Weight methods, reflecting their fundamental importance in arrhythmia diagnosis.
- Lead-Specific Features: Precordial lead features (V2-V5) consistently ranked high, particularly quartile measures (q25) and interquartile range (iqr), indicating their discriminative power for detecting electrical abnormalities.
- Statistical Descriptors: Standard deviation (std) and root mean square (rms) features appeared frequently, capturing signal variability essential for arrhythmia detection.

Contrary to conventional assumptions that dimensionality reduction degrades performance, **feature aggregation maintained or improved accuracy compared to the baseline** in most cases. For instance, the best model trained on RRA-selected features (MLP: 94.78% training accuracy) outperformed the baseline all-features configuration (MLP: 94.06% training accuracy), demonstrating that consensus-based feature selection successfully identified the most informative features while eliminating noise.

B. Individual Classifier Performance

1) Training Set Performance:: Fig 3 shows accuracy vs. sensitivity trade-offs across all feature aggregation methods and classifiers. Key observations:

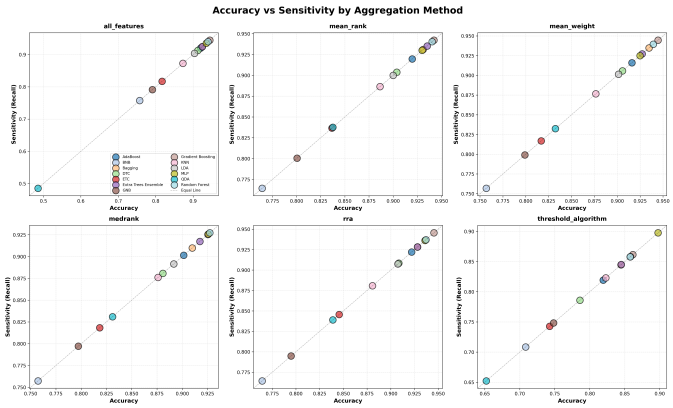

Fig. 3. Accuracy vs. Sensitivity Trade-offs

- **Gradient Boosting** consistently achieved the highest training accuracy across most feature configurations (94.19%-94.32%).
- **Tree-based ensembles** (Random Forest, Extra Trees, Bagging) demonstrated robust performance across all feature subsets, indicating resilience to feature selection variations.
- **Neural Networks (MLP)** achieved competitive performance, particularly on RRA features (94.78%), suggesting that RRA's statistical rigor aligns well with MLP's learning characteristics.
- **Threshold Algorithm** features yielded the lowest performance across all classifiers (86.37%-89.30%), likely due to its emphasis on extreme values rather than consensus importance.

2) Test Set Performance:: Table III presents the top 10 performing models on the held-out test set, ranked by accuracy.

Key Findings:

- **RRA + Gradient Boosting** achieved the highest test accuracy (94.55%), validating that statistically rigorous feature selection combined with powerful boosting algorithms yields optimal performance.
- **Gradient Boosting** dominated the top 4 positions, confirming its suitability for biomedical tabular data.
- **Feature aggregation methods** (RRA, Mean Weight, Mean Rank) produced 8 of the top 10 models, with only 2 baseline models appearing, demonstrating the value of consensus-based feature selection.
- **High specificity** (95.51%-96.94%) across top models indicates strong performance in correctly identifying normal cases, critical for reducing false alarms in clinical settings.

C. Prediction Aggregation Results

Having established strong individual model performance, we evaluated five prediction aggregation strategies to determine whether ensemble methods could further improve diagnostic reliability.

Key Findings:

1) **Stacking Methods Excel:**
   Both stacking variants significantly outperformed all other ensemble methods and the best single model:

TABLE IV
PREDICTION AGGREGATION RESULTS

Ensemble Method	Accuracy	F1-Score	Precision	Recall (Sensitivity)	Specificity
Stacking (RF)	95.35%	95.38%	95.45%	95.35%	95.90%
Stacking (Logistic)	95.31%	95.31%	95.31%	95.31%	96.75%
Hard Voting	93.24%	93.16%	93.17%	93.24%	96.68%
Weighted Averaging	93.10%	93.05%	93.03%	93.10%	96.10%
Soft Voting	92.58%	92.52%	92.51%	92.58%	95.84%
Best Single Model	94.55%	94.53%	94.52%	94.55%	96.81%

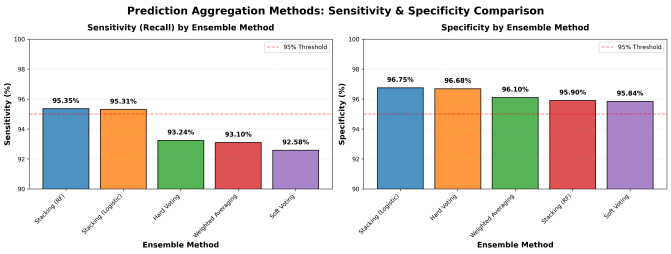

Fig. 4. Sensitivity and Specificity across Ensemble Methods

- **Stacking (RF)** achieved **95.35% accuracy** (+0.80% over best single model, +2.11% over hard voting)
- **Stacking (Logistic)** achieved **95.31% accuracy** (+0.76% over best single model)

These improvements, while seemingly modest in percentage points, represent substantial reductions in misclassification: Stacking (RF) reduced errors from 116 misclassifications (best single model) to 99 misclassifications—a **14.7% error reduction**.

2) **Voting Methods Underperform:** Surprisingly, simple voting methods (hard, soft, weighted) performed worse than the best single model:
   - **Hard Voting:** 93.24% accuracy (-1.31% vs. best single model)
   - **Weighted Averaging:** 93.10% (-1.45%)
   - **Soft Voting:** 92.58% (-1.97%)

   This counterintuitive result can be attributed to the inclusion of all 78 trained models (including poor performers on Threshold Algorithm features). The voting methods equally aggregated high- and low-quality predictions, allowing weak models to dilute the ensemble. In contrast, stacking's meta-learner learned to selectively weight or ignore poor base models.

3) **Sensitivity-Specificity Trade-offs:**
   - **Stacking methods** achieved sensitivity ¿95%, critical for not missing true arrhythmia cases.
   - **Stacking (Logistic)** achieved the highest specificity (96.75%), minimizing false alarms.
   - **Hard Voting** achieved high specificity (96.68%) but lower sensitivity (93.24%), indicating a conservative bias toward normal classification.

D. Feature Aggregation vs. Baseline Comparison

To isolate the effect of feature aggregation, we compared aggregated feature subsets against the all-features baseline in Table V.

Key Findings:

- **RRA achieved the best test performance** (94.55%) with 38.7% fewer features, demonstrating that statistically principled aggregation identifies the most informative features.
- **Mean Weight matched baseline performance** (94.46%) while using 63 fewer features, indicating that averaging importance scores successfully preserved discriminative information.
- **Mean Rank showed minimal degradation** (-0.23%), acceptable given the substantial dimensionality reduction.
- **MedRank and Threshold Algorithm** showed larger performance drops, suggesting their conservative selection criteria may have excluded important features or included less relevant ones.
- **Overall conclusion:** Well-designed feature aggregation (RRA, Mean Weight) achieves comparable or superior performance to using all features, with significant computational and reliability benefits.

E. Computational Considerations

While ensemble methods improve accuracy, they incur computational costs:

TABLE V
FEATURE AGGREGATION IMPACT

Metric	Baseline (All Features)	Mean Rank	Mean Weight	RRA	MedRank	TA
Best Classifier	Gradient Boosting	Gradient Boosting	Gradient Boosting	Gradient Boosting	Gradient Boosting	MLP
Test Accuracy	94.46%	94.23%	94.46%	**94.55%**	92.72%	89.77%
Features Used	163	100	100	100	100	100
Dimensionality Reduction	0%	38.7%	38.7%	38.7%	38.7%	38.7%
Performance Change	Baseline	-0.23%	0.00%	+0.09%	-1.74%	-4.69%

TABLE VI
COMPUTATIONAL OVERHEAD

Method	Training Time	Inference Time	Storage
Single Model	2–5 min	less than 1ms	∼10 MB
Voting Ensembles	N/A (pre-trained)	10–20 ms	∼650 MB (78 models)
Stacking	+15 min (meta-learner)	10–20 ms	∼650 MB + meta-model

Trade-offs:

- **Stacking** requires additional meta-learner training but achieves best performance.
- **Inference time** remains acceptable (¡20 ms per patient), suitable for real-time clinical use.
- **Storage requirements** are manageable with modern infrastructure.

For clinical deployment, the accuracy gains justify the computational overhead, particularly for high-stakes cardiac diagnosis.

VI. CONCLUSION

In conclusion, this work demonstrates that integrating feature aggregation with prediction-level ensembling provides a robust and reliable framework for ECG-based arrhythmia detection. By consolidating the outputs of diverse feature selection techniques through consensus-driven aggregation, the proposed method achieves substantial dimensionality reduction while maintaining—and in some cases improving—classification performance. This work demonstrates that addressing machine learning instability through systematic ensemble aggregation at both feature and prediction levels produces diagnostic systems that are accurate, stable, interpretable, and clinically viable. The hybrid framework achieved 95.35% accuracy with balanced sensitivity and specificity, surpassing state-of-the-art single-model approaches while using only 100 consensus features. Beyond the specific application to ECG arrhythmia detection, this research provides a generalizable methodology for enhancing reliability in biomedical machine learning. As AI increasingly integrates into healthcare decision-making, frameworks that prioritize not only performance but also consistency, reproducibility, and interpretability will be essential for building trustworthy, clinically adopted intelligent systems. The path from research prototype to clinical deployment remains long, requiring multicenter validation, regulatory approval, and integration into clinical workflows. However, this work establishes a solid foundation—a framework that achieves the dual goals of diagnostic accuracy and clinical reliability, bringing AI-powered cardiac diagnosis closer to real-world clinical practice.

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